VOLTA: An Open-Source AI-Powered Engineering Assistant for Electrical and Computer Engineering Education and Practice

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Abstract

This paper presents VOLTA (Versatile Open-source LLM for Technical Assistance), a free, locally-hosted AI assistant designed specifically for electrical and computer engineering (ECE) tasks. Unlike general-purpose large language model (LLM) chatbots, VOLTA integrates domain-specific system prompting, sandboxed Python code execution, and a multi-layer Retrieval-Augmented Generation (RAG) architecture to provide expert-level responses across circuit analysis, PLC/SCADA programming, signal processing, control systems, and 3D/PCB design guidance. Built entirely on open-source tools—including Ollama, DeepSeek-R1, Streamlit, and LlamaIndex—VOLTA operates offline with zero ongoing cost, ensuring data privacy and accessibility in resource-constrained environments. Evaluation through domain-specific prompts demonstrates that VOLTA successfully solves Thevenin equivalent circuits, generates IEC 61131-3 compliant PLC ladder logic, plots Bode diagrams, and provides component-level design guidance. The system architecture, implementation details, and preliminary results are presented, along with a roadmap for Phase 2 RAG integration. VOLTA is released as open-source software under the MIT License.

Keywords: large language models, electrical engineering AI, retrieval-augmented generation, PLC programming, open-source, Streamlit, Ollama, DeepSeek-R1, engineering education

I. INTRODUCTION

The rapid advancement of large language models (LLMs) has opened new possibilities for domain-specific AI assistants capable of supporting complex technical reasoning. In the field of electrical and computer engineering (ECE), practitioners routinely engage with a diverse set of tasks—ranging from circuit simulation and control system design to PLC programming, SCADA configuration, and embedded system development. Traditionally, these tasks require proficiency in specialized software such as MATLAB, KiCad, OpenPLC, and various simulation environments, each with steep learning curves.

Existing general-purpose AI tools such as ChatGPT and Google Gemini offer broad knowledge but lack the domain-specific depth, offline capability, and tool integration required for serious ECE work. Commercial engineering AI solutions, where they exist, are typically expensive and proprietary. This gap motivates the development of VOLTA—a free, locally-hosted, open-source AI engineering assistant built specifically for ECE practitioners and students.

VOLTA (Versatile Open-source LLM for Technical Assistance) addresses this gap by combining three key components: (1) a carefully engineered system prompt that configures the LLM to behave as an expert electrical engineer, (2) a sandboxed Python code executor that allows the AI to write and run SciPy, NumPy, SymPy, and Matplotlib code in real-time, and (3) a planned Retrieval-Augmented Generation (RAG) pipeline that grounds responses in real technical references such as IEEE standards, PLC manufacturer manuals, and open-access textbooks.

This paper makes the following contributions:

1. Design and implementation of a domain-specific AI assistant for ECE tasks using entirely free and open-source components.
2. A multi-layer architecture that integrates local LLM inference, sandboxed code execution, and streaming response delivery through a web-based interface.

3. Preliminary evaluation demonstrating successful resolution of representative ECE problems including circuit analysis, PLC code generation, and control system visualization.
4. Open-source release of the full codebase under the MIT License, enabling reproducibility and community extension.

II. RELATED WORK

Several prior works have explored the use of AI and LLMs in engineering education and practice. Brown et al. [1] demonstrated that GPT-3 exhibits broad reasoning capabilities across STEM domains, though without domain-specific grounding. Subsequent work by Wei et al. [2] introduced chain-of-thought prompting, which significantly improved mathematical and logical reasoning in LLMs—a technique leveraged in VOLTA's system prompt design.

In engineering education, AI tutoring systems have been studied extensively. Graesser et al. [3] showed that intelligent tutoring systems can match human tutors for structured problem-solving tasks. However, existing systems rarely extend to the full breadth of ECE—encompassing circuit simulation, PLC programming, and embedded systems simultaneously.

The concept of Retrieval-Augmented Generation (RAG), introduced by Lewis et al. [4], enables LLMs to ground their responses in external knowledge bases, dramatically reducing hallucination on domain-specific queries. RAG has since been applied in medical [5], legal [6], and scientific [7] domains but remains underexplored in electrical engineering contexts.

Locally-hosted LLMs have gained significant traction with the release of models such as LLaMA [8], Mistral [9], and DeepSeek-R1 [10]. Tools like Ollama [11] have simplified local deployment, enabling privacy-preserving, offline AI inference on consumer hardware. VOLTA builds upon this ecosystem to deliver an accessible, cost-free solution.

To the best of the authors' knowledge, VOLTA represents the first open-source, locally-hosted AI assistant purpose-built for the full spectrum of ECE engineering tasks, combining domain-specific prompting, code execution, and RAG in a single integrated system.

III. SYSTEM ARCHITECTURE

VOLTA is designed as a four-layer architecture, illustrated conceptually below. Each layer serves a distinct function in transforming a user's engineering query into an expert-level, actionable response.

A. Layer 1: Foundation LLM

The core reasoning engine is DeepSeek-R1 7B, a state-of-the-art open-source model renowned for its performance on STEM and mathematical reasoning benchmarks. The model is served locally via Ollama, which provides an OpenAI-compatible REST API at http://localhost:11434. This design choice ensures complete data privacy—no queries are sent to external servers—and eliminates API costs.

VOLTA supports hot-swapping between any locally available Ollama model through the sidebar interface, allowing users to trade response speed for capability (e.g., switching to deepseek-r1:1.5b on RAM-constrained machines).

B. Layer 2: Domain Knowledge Engineering

A critical differentiator between VOLTA and a generic chatbot is its engineered system prompt. The system prompt configures the LLM to adopt the persona of a senior electrical engineer with expertise spanning 20+ domains. Key behavioral directives include:

- Always ask clarifying questions before proceeding (e.g., voltage level, number of phases, safety classification).
- Show step-by-step mathematical derivations with intermediate results.
- Cite relevant standards (IEC, IEEE, NFPA) where applicable.
- Recommend specific components with part numbers when performing design tasks.

- Generate executable Python code for numerical or simulation tasks.

C. Layer 3: Tool Integration and Code Execution

VOLTA includes a sandboxed Python execution module (utils/code_executor.py) that automatically detects, extracts, and runs Python code blocks within AI responses. The sandbox supports NumPy, SciPy, SymPy, and Matplotlib, enabling real-time numerical computation and plot generation within the chat interface.

This capability allows VOLTA to function as a computational notebook—the AI writes the code and the system executes it, displaying results and figures inline. The sandbox is isolated from the system Python environment to prevent unintended side effects.

D. Layer 4: Web Interface

The user interface is built with Streamlit, a Python-native web framework that enables rapid development of interactive data applications. The interface provides a conversational chat panel with streaming token-by-token response delivery, a sidebar for model configuration and Ollama status monitoring, an example prompt library with 15+ pre-built ECE queries, session export to Markdown, and temperature control for balancing precision and creativity.

IV. IMPLEMENTATION

A. Technology Stack

Table I summarizes the complete open-source technology stack used in VOLTA Phase 1, along with the commercial equivalent each tool replaces.

Component	VOLTA (Free)	Commercial Equivalent
LLM Inference	DeepSeek-R1 7B via Ollama	OpenAI GPT-4, Claude API
Math & Simulation	Python + NumPy + SciPy + SymPy	MATLAB
Circuit Simulation	Ngspice / LTspice	PSpice (Cadence)
Control Systems	Python-control library	MATLAB Control Toolbox
PCB Design	KiCad	Altium Designer
3D Modeling	FreeCAD / OpenSCAD	SolidWorks, AutoCAD
PLC Programming	OpenPLC Runtime (IEC 61131-3)	TIA Portal (Siemens)
SCADA	ScadaBR / OpenSCADA	Wonderware, Ignition
Web Interface	Streamlit	Custom / Proprietary
RAG Pipeline (Ph.2)	LlamaIndex + ChromaDB	Azure AI Search

TABLE I. VOLTA TECHNOLOGY STACK AND COMMERCIAL EQUIVALENTS

B. Installation and Deployment

VOLTA is designed for rapid deployment on any Windows, macOS, or Linux system with Python 3.8+ and at least 8GB of RAM. The installation procedure requires five commands:

```
pip install -r requirements.txt
ollama pull deepseek-r1:7b
ollama serve
python -m streamlit run app.py
```

The application launches at http://localhost:8501 and requires no cloud credentials or ongoing subscription fees. The DeepSeek-R1 7B model download is approximately 4.7 GB and is performed once.

C. Session Management and Export

VOLTA maintains a full conversation history within each session, enabling multi-turn engineering dialogues. Users can export the complete session as a structured Markdown document, suitable for engineering notebooks, lab reports, or design documentation. The system stores 15+ pre-built example prompts covering the full ECE domain spectrum.

V. EVALUATION AND RESULTS

To assess VOLTA's capability across its intended domain, a set of representative ECE problems was used to evaluate response quality during Phase 1 testing. Evaluation was conducted qualitatively, assessing correctness, completeness, and practical utility of responses.

A. Circuit Analysis

VOLTA was prompted to calculate the Thevenin equivalent of a 12V source with 4Ω and 6Ω resistors. The system provided a complete step-by-step derivation, correctly identifying the parallel combination for Thevenin resistance (R_{th} = 2.4Ω) and the open-circuit voltage (V_{th} = 12V). A Python code block was generated using SymPy and automatically executed, confirming the analytical result numerically.

B. PLC Ladder Logic Generation

A prompt requesting IEC 61131-3 compliant ladder logic for a motor starter with forward/reverse control was submitted. VOLTA generated syntactically correct structured text (ST) code, included appropriate interlocking contacts to prevent simultaneous forward/reverse activation, and added inline comments referencing relevant IEC 61131-3 clauses.

C. Control Systems Visualization

VOLTA was asked to plot the Bode diagram for a second-order system with natural frequency ω_n = 10 rad/s and damping ratio $\zeta = 0.5$. The system generated complete Python code using the scipy.signal and matplotlib libraries, which was automatically executed, producing magnitude and phase plots with correctly labeled axes, dB scale, and phase in degrees.

D. Response Latency

On a test machine equipped with an Intel Core i7 processor and 16GB RAM, the first response token appeared within 20–30 seconds during model warm-up. Subsequent responses in the same session were delivered in 8–15 seconds for typical engineering queries. Response latency is primarily constrained by the local hardware rather than VOLTA's architecture.

E. Limitations

Phase 1 VOLTA relies solely on the LLM's parametric knowledge without grounding in external references. This introduces the risk of hallucination on highly specific queries (e.g., exact component specifications, obscure IEC sub-clauses). The planned Phase 2 RAG pipeline addresses this by connecting VOLTA to a curated knowledge base of IEEE standards, open-access textbooks, and manufacturer datasheets.

VI. PHASE 2: RAG KNOWLEDGE BASE

The Phase 2 architecture introduces a Retrieval-Augmented Generation pipeline that transforms VOLTA from a smart chatbot into a reference-grade engineering expert. The pipeline comprises four stages:

5. Document Ingestion: IEEE/IEC standards, open-access textbooks (Chapman's Electric Machinery, Nilsson's Circuits), PLC manufacturer manuals (Siemens, Allen-Bradley), MATLAB/Octave documentation, and Digi-Key/Mouser datasheets are parsed using Unstructured.io.
6. Embedding and Indexing: Documents are chunked and embedded using a local embedding model, stored in ChromaDB—a free, open-source vector database.

7. Retrieval: At query time, LlamaIndex retrieves the top-k most semantically relevant document chunks and injects them into the LLM context window.
8. Grounded Generation: The LLM generates responses citing specific retrieved passages, with source attribution displayed in the VOLTA interface.

This architecture enables VOLTA to answer highly specific queries such as transformer sizing per IEC 60076, motor protection coordination per IEEE Std 242, and S7-1200 PLC addressing conventions—with direct citations.

VII. CONCLUSION

This paper presented VOLTA, a free, open-source, locally-hosted AI assistant for electrical and computer engineering. By combining a domain-engineered system prompt, sandboxed Python code execution, and a Streamlit-based web interface with the DeepSeek-R1 7B model via Ollama, VOLTA delivers expert-level ECE assistance at zero cost with complete data privacy.

Preliminary evaluation demonstrates that VOLTA successfully addresses circuit analysis, PLC programming, and control system visualization tasks. The planned Phase 2 RAG integration will further enhance accuracy by grounding responses in curated technical references.

VOLTA is released as open-source software under the MIT License and is publicly available at https://github.com/brijeshpatel271/volta-elec-ai. Future work includes Phase 2 RAG deployment, integration of KiCad schematic generation, multi-user support, and a quantitative evaluation benchmark covering the full ECE domain taxonomy.

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